

(65) Context Free Language

A Context Free Language is a language generated by some Context Free Grammar.

The set of all CFL is identical to the set of languages accepted by Pushdown Automata

CFG: is defined by 4 tuples as $G = \{V, \Sigma, S, P\}$ where:

V = Set of variable or Non-Terminal symbols

Σ = Set of Terminal symbols

S = start symbol

P = production Rule. $\longrightarrow A \rightarrow \alpha$

where, $\alpha \in \{V \cup \Sigma\}^*$ and

$A \in V$

Example: $a^n b^n$

$G = \{ \{S, A\}, \{a, b\}, (S \rightarrow aAb, A \rightarrow aAb | \epsilon) \}$

$$S \rightarrow aAb$$

$$\rightarrow aaAbb \quad (\text{by } A \rightarrow aAb)$$

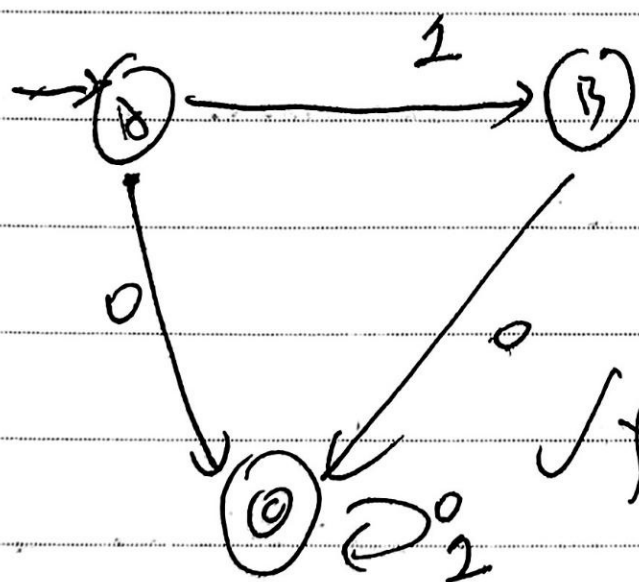
$$\rightarrow aaaaAbbb$$

$$\rightarrow aaaa bbb \quad (\text{by } A \rightarrow \epsilon)$$

$$\rightarrow a^3 b^3 \Rightarrow a^n b^n$$

(66) Regular Languages and Finite Automata

Problem - 1



which False

✓ 1. Complement of $L(A)$ is context free

✓ 2. $L(A) = L((11^*0^*1)^*)$

3. For the Language accepted by A , A is the minimal DFA

4. A accepts all strings